

Mehdi Maxime Bakkali

Senior Data & AI Architect | Lead Software Engineer | Ex-CTO & Founder (8 years of experience)

Location: Paris, France | Remote Available | **Phone:** +33 6 61 34 62 70 | **Email:** mehdi.bakkali.94@gmail.com | **LinkedIn:** [linkedin.com/in/mbakkali](https://www.linkedin.com/in/mbakkali) | **Background:** Engineering (INSA Lyon) + Business (X-HEC Entrepreneurs)

PROFILE

Expert in **Data Engineering and Applied AI** with 8 years of experience. Hybrid profile combining strategic vision as an **Ex-CTO** with technical excellence. Specialist in the complete product lifecycle: from cloud architecture (AWS/GCP) to the deployment of AI agents and complex interfaces using Angular. Pragmatic, ROI-driven approach with a proven track record in scaling data platforms and delivering measurable business outcomes. Co-founder and former CTO of Upfund, a French startup specializing in data & AI systems for commercial real estate.

TECHNICAL SKILLS

- **AI Coding & Productivity:** Expert user of next-gen AI development tools including **Cursor**, **Claude Code**, Replit, Supabase and **LiteLLM** for accelerated delivery and agentic workflows.
- **Data & AI:** Expert in Python (FastAPI, Pandas, Polars) (6 years); Proficient in SQL, PostGIS, and DuckDB (6 years); Advanced in RAG/LangChain and AI Agents; Geospatial Expertise (DeckGL, Kepler).
- **Frontend & Web:** Advanced **TypeScript (8 years)**; Expert in **React** and **Angular**; **NodeJS (3 years)** (Express, Serverless); Data Visualization (DeckGL, Dash).
- Backend skills : Python (6 years) Node JS (3 years) Java (4 years)
- **Cloud & Infrastructure:** Proficient in AWS (Lambda, S3, EC2, Athena, Cognito) and GCP (Vertex AI, Cloud Run, Cloud SQL); Docker and Kubernetes; Advanced DevOps with CI/CD (GitHub Actions, BitBucket Pipelines).

PROFESSIONAL EXPERIENCE

UPFUND — CEO & Co-founder

March 2021 – Present | Paris

- **Technical Scale:** Designed and deployed a geospatial data platform processing +100 daily data sources using Prefect, DuckDB, and PostGIS for commercial real estate analytics.
- **Frontend Excellence:** Developed advanced geospatial visualization interfaces (React/Angular + DeckGL) enabling intuitive exploration of complex spatial datasets.
- **Leadership:** Built and managed a high-performing engineering team of 8 (Data, Backend, Product), establishing technical standards and mentoring across the full stack.
- **Generative AI:** Architected proprietary RAG systems for automated analysis of complex real estate data, enabling intelligent document processing and insight generation at scale.
- **Business Impact:** Secured \$1.5M in funding and delivered enterprise-grade projects for strategic clients including BNP Real Estate, CBRE, L'Oréal, and the French Government.
- **Cloud Architecture:** Designed hybrid multi-cloud infrastructure (AWS & GCP) ensuring high availability and scalability for AI workloads and real-time data processing.

JCDECAUX — Freelance Software Engineer

Feb. 2020 – Feb. 2021

- **Data Portal:** Built an enterprise analytical dashboard to unlock insights from advertising data warehouses, processing millions of display panel logs daily.
- **User Experience:** Designed and developed interactive data visualization interfaces (Angular) enabling marketing teams to analyze campaign performance in real-time.
- **Efficiency Gains:** Created Python Dash modules for advanced analytics and custom reporting, reducing manual analysis time by 70%.
- **Performance Optimization:** Architected a scalable Java Quarkus backend with distributed queries via AWS Athena, improving query performance by 5x.

- **Business Impact:** Transformed raw advertising data into actionable insights, directly supporting strategic decision-making for executive leadership.
- **References:** Habiba Bouchareb (PM) | Hatim Tachi (Tech Lead) - 07 82 03 15 62.

PERNOD RICARD — Freelance Full Stack Developer

Jan. 2019 – June 2020 (1.5 years)

- **Magellan / Data Portal:** Developed a data mapping and cataloging application managing 10,000+ data sources for the Pernod Ricard group.
- **Architecture:** Developed V1 using serverless architecture (NodeJS Lambda) ensuring cost efficiency and automatic scalability.
- **Design System:** Defined a comprehensive design system and reusable component library, accelerating future development cycles.
- **DevOps Excellence:** Implemented automated CI/CD pipelines via BitBucket and AWS SAM, reducing deployment time by 80%.
- **Cross-functional Collaboration:** Partnered with BI/Data teams to design robust data catalog models, improving data discoverability across the organization.
- **Reference:** Erwan Raulo (Tech Lead) - +33 6 21 49 43 36.

GRT GAZ — Freelance Full Stack Developer

Jan. 2018 – June 2019 (1.5 years)

- **Field Operations Digitalization:** Developed web/mobile applications for 2,000+ field technicians at the French gas network operator.
- **Cross-platform Development:** Built multi-platform apps using Angular/Ionic/Capacitor, ensuring consistent experience across web, iOS, and Android.
- **API Architecture:** Designed NodeJS/Express REST APIs enabling seamless integration with legacy internal systems.
- **Scale & Maintenance:** Maintained and evolved 20+ business-critical applications, ensuring 99.9% uptime for field operations.
- **Cloud & Agile:** Led AWS cloud deployment and CI/CD automation within an Agile/Scrum framework.
- **Reference:** Hamza Guitouni (Project Manager) - +33 6 35 66 13 71.

SOCIÉTÉ GÉNÉRALE CIB — Trading Floor Developer (Permanent)

Sept. 2016 – Feb. 2018

- **High-stakes Environment:** Supported front-office traders on the equity derivatives desk in fast-paced market conditions.
- **Automation:** Developed Java microservices for automated pricing and reporting, reducing manual intervention and operational risk.
- **Real-time Visualization:** Built Angular dashboards for real-time PnL tracking and risk exposure monitoring, critical for trading decisions.
- **Risk Management:** Created Python/VBA scripts for complex risk calculations and data quality validation, ensuring regulatory compliance.
- **Regulatory Compliance:** Implemented front-office workflows compliant with MiFID II European financial regulations.
- **Reference:** Julien Messelier (IT Desk Manager) - +33 6 81 16 30 57.

EDUCATION

- **MSc X-HEC Entrepreneurs** | HEC Paris & École Polytechnique (2020 - 2021)
 - Specialization: technology entrepreneurship, AI & data.
- **Telecommunications & IT Engineering Degree** | INSA Lyon (2012 - 2017)
 - Specialization: algorithms, data, web & networks.

LANGUAGES & INTERESTS

- **Languages:** French (Native) , English (Bilingual) , Arabic (Native) , Spanish (B1) , German (B1).
- **Interests:** Volleyball - National Level N2 (competitive sports background); International travel and cultural adaptability; Generative AI active exploration; Process automation enthusiast.